

Causal DAG

METIMMOX-1 Economic Evaluation

Ben Geisler

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1 Introduction

This report presents the directed acyclic graph (DAG) for the METIMMOX-1 cost-effectiveness analysis. The DAG encodes the assumed causal structure between patient characteristics, biomarkers, treatment, intermediate outcomes, and survival endpoints. It was specified in DAGitty format and visualised using the **ggdag** package in R.

Three node types are distinguished:

- **Exposure** (T): the randomised treatment (alternating FLOX + nivolumab vs FLOX alone)
- **Outcomes** (PFS, OS): the clinical endpoints of interest
- **Latent** (U): unmeasured patient characteristics that act as common causes of biomarkers and/or survival

2 Node Definitions

Table 1: DAG node definitions

Node	Display	Type	Description
Age	Age	Covariate	Patient age at baseline
Sex	Sex	Covariate	Patient sex
U	U	Latent	Unmeasured prognostic factors (latent)
TMB_BRAF	TMB/BRAF	Biomarker	Combined biomarker: TMB ≥ 9 mut/MB or BRAF mutation
CRP	CRP	Biomarker	C-reactive protein (positive: CRP < 5 mg/L)
T	T	Exposure	Treatment: alternating FLOX + nivolumab vs FLOX alone
TxTMB	T x TMB	Effect modifier	Interaction between treatment and TMB/BRAF status
TxCRP	T x CRP	Effect modifier	Interaction between treatment and CRP status
TLR	TLR	Mediator	Tumour lesion reduction (radiological response)
PFS	PFS	Outcome	Progression-free survival
OS	OS	Outcome	Overall survival

3 Causal Structure

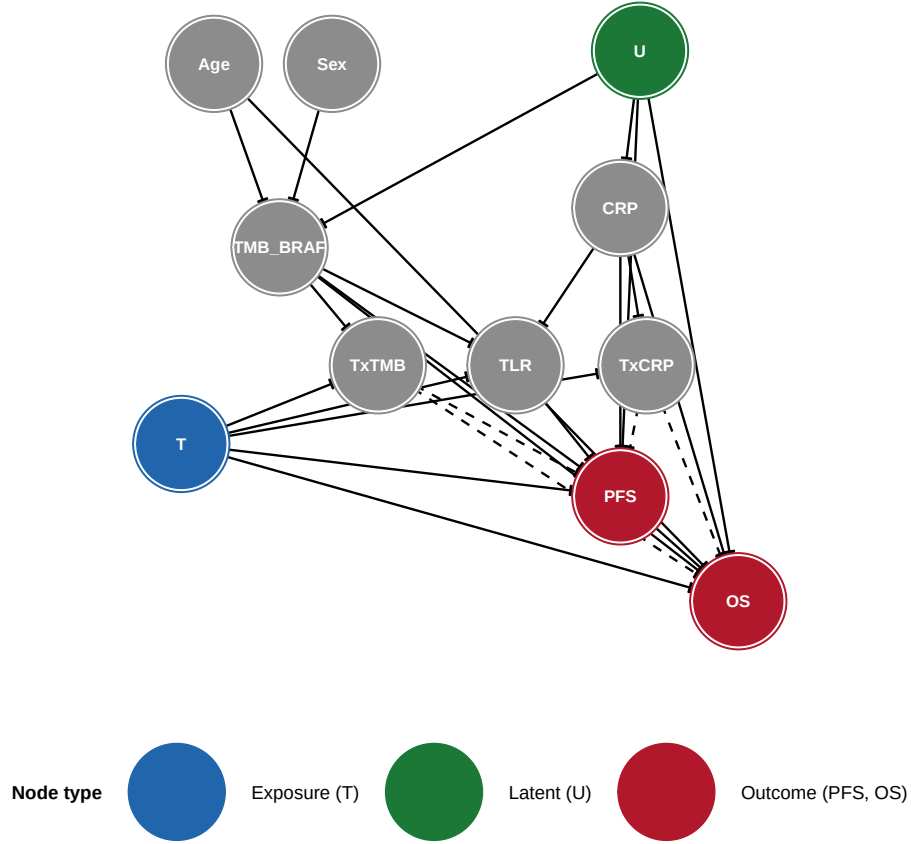


Figure 1: Causal DAG for the METIMMOX-1 analysis. Node colour indicates role: blue = exposure (T), red = outcomes (PFS, OS), green = latent variable (U), grey = covariates and intermediate variables. Solid arrows denote assumed causal effects; dashed arrows indicate the hypothesised interaction effects of T x TMB/BRAF and T x CRP on PFS and OS. Node abbreviations: T = Treatment; TxTMB = Treatment x TMB/BRAF interaction; TxCRP = Treatment x CRP interaction; TLR = Tumour Lesion Reduction; U = Unmeasured confounders.

4 Causal Analysis

Because T is a randomised exposure with no parents in the DAG, there are no backdoor paths from T to PFS or OS. The sections below confirm this using dagitty's formal algorithms and list the implied conditional independencies encoded by the DAG.

4.1 Adjustment Sets

T -> PFS: Minimal adjustment sets: {}

T -> OS: Minimal adjustment sets: {}

4.2 Implied Conditional Independencies

Table 2: Conditional independencies implied by the causal DAG

Implied conditional independence ($X \perp\!\!\!\perp Y \mid \{Z\}$)
Age $\perp\!\!\!\perp$ CRP
Age $\perp\!\!\!\perp$ Sex
Age $\perp\!\!\!\perp$ T
Age $\perp\!\!\!\perp$ TLR $\mid \{\text{CRP, TMB_BRAF}\}$
Age $\perp\!\!\!\perp$ TxCRP
Age $\perp\!\!\!\perp$ TxTMB $\mid \{\text{TMB_BRAF}\}$
CRP $\perp\!\!\!\perp$ Sex
CRP $\perp\!\!\!\perp$ T
CRP $\perp\!\!\!\perp$ TxTMB $\mid \{\text{TMB_BRAF}\}$
Sex $\perp\!\!\!\perp$ T
Sex $\perp\!\!\!\perp$ TLR $\mid \{\text{CRP, TMB_BRAF}\}$
Sex $\perp\!\!\!\perp$ TxCRP
Sex $\perp\!\!\!\perp$ TxTMB $\mid \{\text{TMB_BRAF}\}$
T $\perp\!\!\!\perp$ TMB_BRAF
TLR $\perp\!\!\!\perp$ TxCRP $\mid \{\text{CRP, T}\}$
TLR $\perp\!\!\!\perp$ TxTMB $\mid \{\text{T, TMB_BRAF}\}$
TMB_BRAF $\perp\!\!\!\perp$ TxCRP $\mid \{\text{CRP}\}$
TxCRP $\perp\!\!\!\perp$ TxTMB $\mid \{\text{T, TMB_BRAF}\}$
TxCRP $\perp\!\!\!\perp$ TxTMB $\mid \{\text{CRP, T}\}$

5 Sensitivity Analysis: Excluding Age/Sex $\rightarrow$ TMB/BRAF

This sensitivity DAG removes the **Age** $\rightarrow$ **TMB_BRAF** and **Sex** $\rightarrow$ **TMB_BRAF** edges while keeping the remainder of the structure identical to the primary specification. The primary DAG retains these arrows on biological grounds; this sensitivity analysis shows the downstream implications of omitting them.

5.1 Sensitivity DAG

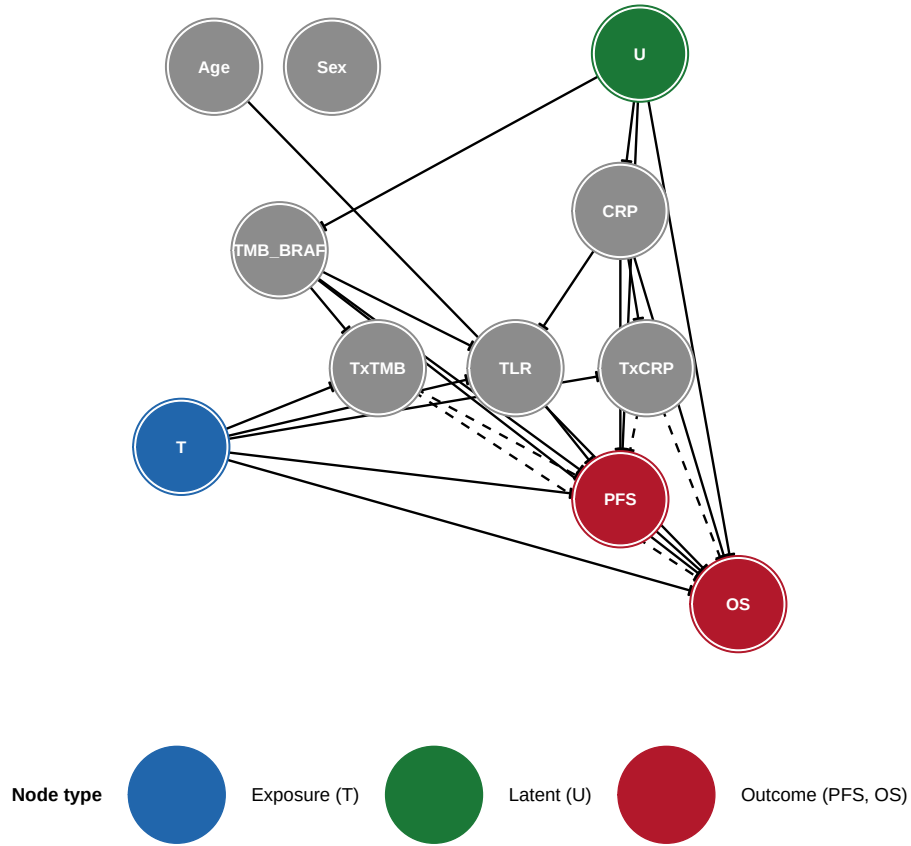


Figure 2: Sensitivity DAG for the METIMMOX-1 analysis, excluding the Age → TMB_BRAF and Sex → TMB_BRAF edges while retaining all other assumed causal relationships. Solid arrows denote assumed causal effects; dashed arrows indicate the hypothesised interaction effects of T x TMB/BRAF and T x CRP on PFS and OS.

5.2 Sensitivity Causal Analysis

The treatment node remains randomised in the sensitivity DAG, so the formal adjustment recommendations for T → PFS and T → OS are unchanged. The main difference is in the implied conditional independencies, which now reflect the omission of direct Age/Sex effects on TMB/BRAF status.

5.2.1 Adjustment Sets

T → PFS: Minimal adjustment sets: {}

T → OS: Minimal adjustment sets: {}

5.2.2 Implied Conditional Independencies

Table 3: Conditional independencies implied by the sensitivity DAG

Implied conditional independence ($X \perp\!\!\!\perp Y \mid \{Z\}$)
Age $\perp\!\!\!\perp$ CRP
Age $\perp\!\!\!\perp$ PFS
Age $\perp\!\!\!\perp$ Sex
Age $\perp\!\!\!\perp$ T
Age $\perp\!\!\!\perp$ TLR
Age $\perp\!\!\!\perp$ TMB_BRAF
Age $\perp\!\!\!\perp$ TxCRP
Age $\perp\!\!\!\perp$ TxTMB
CRP $\perp\!\!\!\perp$ Sex
CRP $\perp\!\!\!\perp$ T
CRP $\perp\!\!\!\perp$ TxTMB $\mid \{TMB_BRAF\}$
OS $\perp\!\!\!\perp$ Sex
PFS $\perp\!\!\!\perp$ Sex
Sex $\perp\!\!\!\perp$ T
Sex $\perp\!\!\!\perp$ TLR
Sex $\perp\!\!\!\perp$ TMB_BRAF
Sex $\perp\!\!\!\perp$ TxCRP
Sex $\perp\!\!\!\perp$ TxTMB
T $\perp\!\!\!\perp$ TMB_BRAF
TLR $\perp\!\!\!\perp$ TxCRP $\mid \{CRP, T\}$
TLR $\perp\!\!\!\perp$ TxTMB $\mid \{T, TMB_BRAF\}$
TMB_BRAF $\perp\!\!\!\perp$ TxCRP $\mid \{CRP\}$
TxCRP $\perp\!\!\!\perp$ TxTMB $\mid \{T, TMB_BRAF\}$
TxCRP $\perp\!\!\!\perp$ TxTMB $\mid \{CRP, T\}$

6 Simplified DAG

The following DAG presents the same causal assumptions in a condensed form intended for a clinical audience. The interaction terms ($T \times TMB/BRAF$ and $T \times CRP$) and the latent common-cause node (U) have been omitted; their roles are described in the text. Dashed arrows from T to PFS and OS indicate that the magnitude of the treatment effect on survival outcomes is hypothesised to be modified by TMB/BRAF and CRP status — the central question addressed by the biomarker analysis.

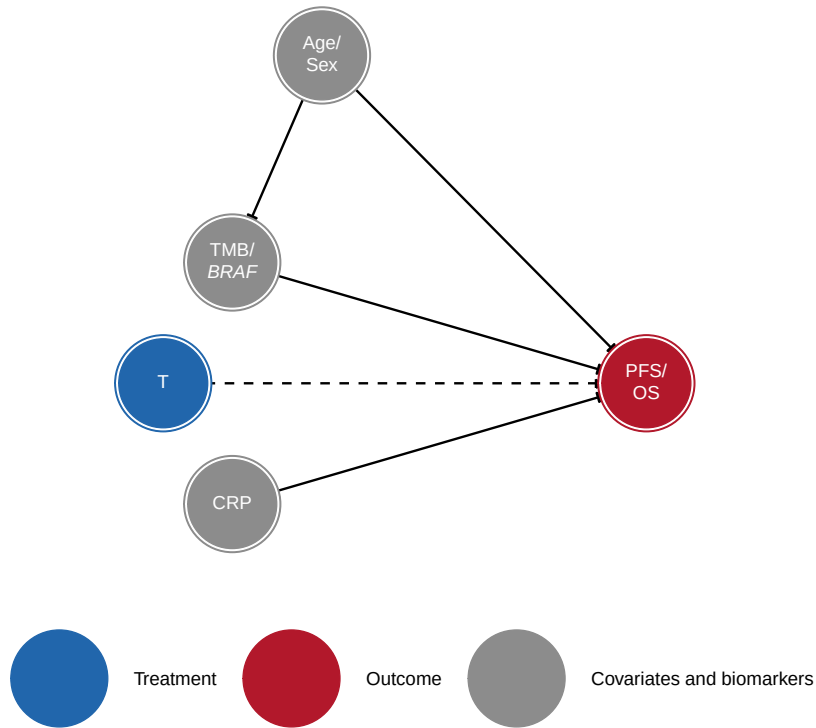


Figure 3: Figure 3. Simplified causal DAG for METIMMOX-1. Solid arrows denote assumed causal effects. The dashed arrow indicates that the treatment effect on survival is hypothesised to be modified by TMB/BRAF and CRP status. Interaction and latent-variable nodes are shown in the full DAG. Abbreviations: CRP = C-reactive protein, OS = overall survival, PFS = progression-free survival, T = treatment, TMB = tumour mutational burden.

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